

DengueCAD — Performance Comparison Report

NMPrediction (Algorithm 1: NMI imputation → GA wrapper + ADT feature selection → stratified k-fold ADT) compared with C4.5, logistic regression (LOR), and SVM-RBF. Based on Rao & Kumar, IEEE TITB 2012.

Part I covers **dengue-related** experiments. Part II covers **non-dengue** multi-disease ARFF benchmarks and the MissForest Iris comparison.

Legend — abbreviations used in this report

MICE means *Multiple Imputation by Chained Equations*: an iterative method that models each incomplete column as a function of the others and cycles until convergence. In DengueCAD it is implemented with scikit-learn's `IterativeImputer` (a MICE-style chained equations imputer).

Abbreviation	Full form / meaning
ADT	Alternating Decision Tree (Freund & Mason); used inside NMPrediction
ARFF	Attribute-Relation File Format (Weka / KEEL datasets)
AUC	Area Under the ROC Curve
C4.5	Quinlan decision-tree learner (entropy / information-gain style; CART stand-in in scikit-learn)
CV	Cross-Validation (here: stratified k-fold)
dengue (train)	Single 70k-row training block from dengue.csv for NMI / NMPrediction
GA	Genetic Algorithm (wrapper search for influential feature subsets)
kNN	k-Nearest Neighbours imputation
LOR	Logistic Regression
MCAR	Missing Completely At Random (synthetic missingness pattern)
MICE	Multiple Imputation by Chained Equations (<code>IterativeImputer</code> in this project)
MissForest	Random-forest-based missing-value imputation (Stekhoven & Bühlmann)
MV	Missing Value(s)
NM	New Methodology / NMPrediction (IEEE TITB 2012 Algorithm 1)
NMI	Non-parametric Missing-value Imputation (indexing / nearest-neighbour method)
NRMSE	Normalized Root Mean Squared Error (imputation error for continuous attributes)
PFC	Proportion of Falsely Classified entries (imputation error for categoricals)
RBF	Radial Basis Function kernel (SVM)

Abbreviation	Full form / meaning
ROC	Receiver Operating Characteristic curve
SE	Sensitivity (true positive rate / recall)
SP	Specificity (true negative rate)
SVM	Support Vector Machine
TITB / JBHI	IEEE Transactions on Information Technology in Biomedicine / Journal of Biomedical Health Informatics
UCI	University of California, Irvine Machine Learning Repository
Weka / KEEL	Open-source ML workbenches; source of the multi-disease ARFF files

Row colours in result tables: **Red** = best-performing model in that comparison (including NM / NMI when it wins); **Green** = NM / NMI / NMPrediction row when it is *not* the best performer.

Part I — Dengue-related performance comparisons

Datasets: synthetic/vendored dengue.csv (70k single train set), Bangladesh dataset.csv, and Dengue_clinical_dataset.csv (Munshiganj clinical/hematology).

1. Vendored dengue.csv — NMPrediction (70k single dataset)

Protocol: first 70,000 rows as one training dataset for NMI / NMPrediction; 10,000 complete-case rows → validation; 20,000 complete-case rows → test (performance). Note: this file has only four binary symptoms and near-chance label signal, so accuracies cluster near 50%.

Table I — Performance (Accuracy / SE / SP / AUC)

Dataset	Method	Accuracy (%)	SE	SP	AUC
dengue	NM	49.85	49.76	49.94	0.5
dengue	C4.5	49.98	49.25	50.71	0.5
dengue	LOR	50.03	49.4	50.67	0.5
dengue	SVM	50.0	50.89	49.1	0.5

Source: results/table1_performance.csv

Table II — Wilcoxon matched-pairs (NM vs baselines)

Dataset	Method	Rank sum (+, −)	p-value
dengue	C4.5	23.0, 32.0	0.695
dengue	LOR	37.0, 18.0	0.375
dengue	SVM	32.0, 23.0	0.695

Source: results/table2_wilcoxon.csv

Table III — Influential features (NM)

Dataset	# Original features	# influential features	Accuracy (%)	features identified
dengue	4	1	49.85	Headache

Source: results/table3_influential_features.csv

Hold-out validation (10k) / test (20k) complete-case

Dataset	Holdout	n	Accuracy (%)	SE	SP	AUC	features
dengue	validation	10000	50.32	51.26	49.41	0.5	Headache
dengue	test	20000	50.39	49.91	50.87	0.5	Headache

Source: results/holdout_metrics.csv

2. Dengue_clinical_dataset.csv (clinical + hematology)

1,018 records (697 dengue-positive / 321 negative). Features include Gender, Age, Platelet Count, WBC, Location, Fever, Duration_of_Fever, Headache, Muscle_Pain, Rash, Vomiting. NM selected Platelet Count, WBC, Fever.

Table I — Performance

Dataset	Method	Accuracy (%)	SE	SP	AUC
Dengue_clinical	NM	98.72	98.85	98.44	0.99
Dengue_clinical	C4.5	97.25	98.13	95.33	0.97
Dengue_clinical	LOR	81.04	80.06	83.18	0.89
Dengue_clinical	SVM	93.71	93.97	93.15	0.98

Source: results/nmprediction_clinical/table1_performance.csv

Table II — Wilcoxon

Dataset	Method	Rank sum (+, -)	p-value
Dengue_clinical	C4.5	15.0, 0.0	0.062
Dengue_clinical	LOR	15.0, 0.0	0.062
Dengue_clinical	SVM	15.0, 0.0	0.062

Source: results/nmprediction_clinical/table2_wilcoxon.csv

Table III — Influential features

Dataset	# Original features	# influential features	Accuracy (%)	features identified
Dengue_clinical	11	3	98.72	Platelet Count, WBC, Fever

Source: results/nmprediction_clinical/table3_influential_features.csv

3. dataset.csv (Bangladesh comprehensive clinical set)

Data caveat: In this file IgG equals Outcome for all 1,000 rows (perfect label leak). Body_Temperature alone also yields 100% majority accuracy. Perfect scores below reflect the dataset structure. A clinical-only run (excluding IgG/NS1) is also shown.

Table I — All features (includes IgG)

Dataset	Method	Accuracy (%)	SE	SP	AUC
dataset.csv	NM	100.0	100.0	100.0	1.0
dataset.csv	C4.5	100.0	100.0	100.0	1.0
dataset.csv	LOR	100.0	100.0	100.0	1.0
dataset.csv	SVM	100.0	100.0	100.0	1.0

Source: results/nmprediction_dataset_csv/table1_performance.csv

Table I — Clinical features only (no IgG / NS1)

Dataset	Method	Accuracy (%)	SE	SP	AUC
dataset.csv (clinical only)	NM	100.0	100.0	100.0	1.0
dataset.csv (clinical only)	C4.5	100.0	100.0	100.0	1.0
dataset.csv (clinical only)	LOR	100.0	100.0	100.0	1.0
dataset.csv (clinical only)	SVM	100.0	100.0	100.0	1.0

Source: results/nmprediction_dataset_csv/table1_performance_clinical.csv

Table III — Influential features (all features run)

Dataset	# Original features	# influential features	Accuracy (%)	features identified
dataset.csv	14	1	100.0	IgG

Source: results/nmprediction_dataset_csv/table3_influential_features.csv

4. Missing-value imputation on dataset.csv (Joint_Pain MVs)

Natural missingness: 491 cells in Joint_Pain. Controlled benchmark: complete cases, 12% MCAR mask, compare NMI vs Mode/Mean, Mean, Median, kNN, MICE.

Imputation accuracy comparison

Method	Accuracy (%)	n_compared	n_match	seconds
NM (NMI)	52.94	408	216	118.29
Mode/Mean	52.94	408	216	0.031
Mean	52.45	408	214	0.003

Method	Accuracy (%)	n_compared	n_match	seconds
Median	52.45	408	214	0.003
kNN	50.74	408	207	0.009
MICE	55.64	408	227	0.146

Source: results/imputation_dataset_csv/imputation_comparison.csv

Part II — Non-dengue performance comparisons

These experiments show that NMI / NMPrediction are **disease-agnostic**: the same pipeline runs on Weka/KEEL ARFF medical datasets and on the canonical MissForest Iris demo (UCI).

5. Multi-disease ARFF benchmark (11 datasets)

Source folder: experiments/datasets/*.arff (appendicitis, bupa, contractions, heart, hepatitis, laryngeal1, lplaudiob, pima, rds, weaning, wisconsin).

Overview — Accuracy (%) by method

Dataset	NM Acc (%)	C4.5 Acc (%)	LOR Acc (%)	SVM Acc (%)
appendicitis	75.47	77.36	80.19	84.91
bupa	62.32	57.39	61.45	59.42
contractions	78.57	78.57	81.63	84.69
heart	85.56	74.44	84.07	84.07
hepatitis	81.29	80.0	78.71	85.16
laryngeal1	82.63	79.34	83.1	85.45
lplaudiob	64.31	55.96	63.77	64.21
pima	70.83	66.41	67.32	68.49
rds	94.12	89.41	92.94	95.29
weaning	86.42	77.81	82.12	84.11
wisconsin	95.71	93.56	97.14	97.0

Pivoted from ALL_table1_performance.csv. NM best on 5/11 datasets in this run.

Table I — Full metrics (Accuracy / SE / SP / AUC)

Dataset	Method	Accuracy (%)	SE	SP	AUC
appendicitis	NM	75.47	85.71	72.94	0.83
appendicitis	C4.5	77.36	52.38	83.53	0.68
appendicitis	LOR	80.19	80.95	80.0	0.85
appendicitis	SVM	84.91	71.43	88.24	0.8

Dataset	Method	Accuracy (%)	SE	SP	AUC
bupa	NM	62.32	56.5	70.34	0.68
bupa	C4.5	57.39	62.5	50.34	0.56
bupa	LOR	61.45	64.0	57.93	0.61
bupa	SVM	59.42	63.0	54.48	0.62
contractions	NM	78.57	75.51	81.63	0.86
contractions	C4.5	78.57	75.51	81.63	0.79
contractions	LOR	81.63	79.59	83.67	0.9
contractions	SVM	84.69	91.84	77.55	0.92
heart	NM	85.56	85.83	85.33	0.91
heart	C4.5	74.44	66.67	80.67	0.74
heart	LOR	84.07	83.33	84.67	0.91
heart	SVM	84.07	85.83	82.67	0.9
hepatitis	NM	81.29	78.86	90.62	0.89
hepatitis	C4.5	80.0	82.11	71.88	0.77
hepatitis	LOR	78.71	76.42	87.5	0.85
hepatitis	SVM	85.16	85.37	84.38	0.89
laryngeal1	NM	82.63	80.3	86.42	0.89
laryngeal1	C4.5	79.34	86.36	67.9	0.77
laryngeal1	LOR	83.1	79.55	88.89	0.88
laryngeal1	SVM	85.45	83.33	88.89	0.91
lplaudiob	NM	64.31	67.28	60.88	0.67
lplaudiob	C4.5	55.96	53.37	58.96	0.56
lplaudiob	LOR	63.77	62.31	65.46	0.67
lplaudiob	SVM	64.21	68.11	59.71	0.67
pima	NM	70.83	70.15	71.2	0.77
pima	C4.5	66.41	51.49	74.4	0.63
pima	LOR	67.32	67.54	67.2	0.74

Dataset	Method	Accuracy (%)	SE	SP	AUC
pima	SVM	68.49	73.51	65.8	0.74
rds	NM	94.12	100.0	88.89	0.97
rds	C4.5	89.41	85.0	93.33	0.89
rds	LOR	92.94	92.5	93.33	0.96
rds	SVM	95.29	95.0	95.56	0.98
weaning	NM	86.42	86.75	86.09	0.92
weaning	C4.5	77.81	78.15	77.48	0.78
weaning	LOR	82.12	84.11	80.13	0.87
weaning	SVM	84.11	89.4	78.81	0.9
wisconsin	NM	95.71	96.68	95.2	0.99
wisconsin	C4.5	93.56	90.04	95.41	0.93
wisconsin	LOR	97.14	98.34	96.51	0.99
wisconsin	SVM	97.0	97.1	96.94	0.99

Source: results/arff_disease_comparison/ALL_table1_performance.csv

Table II — Wilcoxon (NM vs baselines)

Dataset	Method	Rank sum (+, −)	p-value
appendicitis	C4.5	10.0, 0.0	0.125
appendicitis	LOR	1.0, 2.0	1.0
appendicitis	SVM	3.0, 3.0	1.0
bupa	C4.5	10.0, 0.0	0.125
bupa	LOR	12.0, 3.0	0.312
bupa	SVM	10.5, 4.5	0.5
contractions	C4.5	2.0, 4.0	0.75
contractions	LOR	5.0, 10.0	0.562
contractions	SVM	1.0, 5.0	0.5
heart	C4.5	15.0, 0.0	0.062
heart	LOR	12.0, 3.0	0.312
heart	SVM	5.0, 1.0	0.5
hepatitis	C4.5	13.5, 1.5	0.188
hepatitis	LOR	4.5, 1.5	0.75
hepatitis	SVM	4.0, 6.0	1.0
laryngeal1	C4.5	7.0, 3.0	0.625
laryngeal1	LOR	8.0, 2.0	0.375
laryngeal1	SVM	5.0, 10.0	0.625
lplaudiob	C4.5	15.0, 0.0	0.062
lplaudiob	LOR	9.0, 1.0	0.25
lplaudiob	SVM	8.0, 7.0	1.0
pima	C4.5	15.0, 0.0	0.062
pima	LOR	8.0, 7.0	1.0
pima	SVM	11.5, 3.5	0.375
rds	C4.5	2.0, 4.0	1.0
rds	LOR	3.0, 3.0	1.0

Dataset	Method	Rank sum (+, -)	p-value
rds	SVM	0.0, 3.0	0.5
weaning	C4.5	15.0, 0.0	0.062
weaning	LOR	10.0, 0.0	0.125
weaning	SVM	5.0, 1.0	0.5
wisconsin	C4.5	9.0, 1.0	0.25
wisconsin	LOR	0.0, 15.0	0.062
wisconsin	SVM	0.0, 15.0	0.062

Source: results/arff_disease_comparison/ALL_table2_wilcoxon.csv

Table III — Influential feature subsets (NM)

Dataset	# Original features	# influential features	Accuracy (%)	features identified
appendicitis	7	3	75.47	At1, At2, At3
bupa	6	6	62.32	mcvmean, alkphosalkaline, sgptalamine, sgotaspartate, gammagtgamma, drinks
contractions	27	8	78.57	MI1, MI4, MI7, holesize6, holesize8, globalcontrast2, globalcontrast3, globalcontrast5
heart	13	6	85.56	sex, chestpaintype, ElAngina, slope, Majorvessels, thal
hepatitis	19	9	81.29	Sex, Steroid, SpleenPalpable, Spiders, Ascites, Bilirubin, AlbuMin, ProTime, Histology
laryngeal1	16	8	82.63	SD, PPQ, APQ, HNRq, NNE, DH, TNI, NFHE
lplaudiob	7	4	64.31	test, loc, age, d
pima	8	4	70.83	plasmagluucose, Skinthickness, seruminsulin, age
rds	17	13	94.12	Pathology, pretermrupture, age, weight, maturity, APGAR, PHUmbilical, Phressuscitation, BE, Heartrate1, Respiratoryinsufficiency, Crepitation, heartrate2
weaning	17	17	86.42	t, Hb, Ht, Lv, Ru, O, Al, Kr, Lak, FiO, PaO2, PaCO2, PaOFiO2, R, RRs, RRd, RRm
wisconsin	9	6	95.71	ClumpThickness, CellSize, CellShape, EpithelialSize, BareNuclei, BlandChromatin

Source: results/arff_disease_comparison/ALL_table3_influential_features.csv

6. MissForest canonical dataset (UCI Iris) — imputation + prediction

Dataset: UCI Iris (Fisher 1936), the demo set used in the missForest R package (Stekhoven & Bühlmann). Protocol: 20% MCAR on attributes, then impute and classify. Local copy: data/missforest_iris/iris.csv.

Table A — Imputation error (lower NRMSE better)

Method	NRMSE_proxy	Approx match (%)	n_compared	seconds
MICE	0.5322	45.83	120	0.081
MissForest (RF)	0.6335	47.5	120	1.262
NM (NMI)	0.6371	31.67	120	11.006
kNN	0.6776	41.67	120	0.004
Mean	1.0039	21.67	120	0.002
Mode/Mean	1.0039	21.67	120	0.007
Median	1.0267	25.0	120	0.002
MissForest (pkg)	3.1495	10.83	120	0.0

Source: results/missforest_comparison/table_imputation.csv

Table B — Downstream prediction accuracy (%) after imputation

Imputer	ADT	C4.5	LOR	SVM
Complete (no MV)	96.0	95.33	95.33	96.0
MissForest (pkg)	90.0	88.0	93.33	94.0
MissForest (RF)	88.67	92.67	95.33	96.0
NM (NMI)	92.67	90.67	94.67	94.67
Mode/Mean	89.33	85.33	82.0	85.33
Mean	89.33	85.33	82.0	85.33
Median	90.0	86.67	86.0	86.0
kNN	88.0	89.33	92.67	94.0
MICE	94.67	94.0	95.33	96.0

Source: results/missforest_comparison/table_prediction.csv

Generated by DengueCAD scripts/build_comparison_pdf.py. See the **Legend** at the start of this document for all abbreviations (including **MICE** = Multiple Imputation by Chained Equations).